

Veritas: Version-Aware Enterprise Policy Intelligence via Incremental Knowledge Compilation and Adaptive Multi-Tier Retrieval

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Abstract—Enterprise policy repositories undergo continuous revision, requiring historical and active provisions to coexist under strict role-based access. Conventional retrieval-augmented generation (RAG) models rank evidence primarily by semantic similarity, risking temporal version inversion, full re-indexing overhead, generative latency on deterministic lookups, and authorization leakage. We present Veritas, an enterprise policy intelligence system that treats retrieval as a constrained systems problem: a composite eligibility predicate $E(c, u, t_q) = V(c, t_q) \cdot A(c, u)$ enforces date-bounded validity and role authorization prior to evidence admission and reranking. Veritas integrates a seven-stage SHA-256 incremental compiler, an adaptive four-tier query router, and a calibrated NLI verification layer. We evaluate Veritas across two clearly delineated regimes: (1) a frozen 301-query baseline benchmark (commit 61d64b71; 21 policies, 128 chunks), achieving 100.0% authorization correctness (0/301 leaks), 88.07% answered-query version selection accuracy (96/109), and 54.15% deterministic fast-path routing with 29.65 ms median latency; and (2) an expanded 922-query characterization suite (commit 449e69b3; 120 policies, 3,000 chunks, 32 user archetypes), demonstrating zero observed unauthorized evidence leaks across 3,840 evaluations (95% upper bound $< 0.08\%$), 63.4% deterministic/refusal paths, 24.95 $\times$ compiler speedup on 5% amendment deltas across 3,000 chunks (95.0% compute avoided), strict chunk-level Recall@10 of 98.29%, and 1.94 ms end-to-end median latency at enterprise scale. Isotonic calibration achieves 0.041 validation ECE (0.000 on fit set), rising to 0.2954 under runtime distribution shift. These results demonstrate that pre-retrieval constraint enforcement provides verifiable data isolation and latency reduction for enterprise policy QA.

Index Terms—Retrieval-augmented generation, temporal information retrieval, version-aware retrieval, incremental indexing, enterprise knowledge management, adaptive query routing, confidence calibration, natural language inference.

I. INTRODUCTION

ENTERPRISE governance, corporate administration, information security, and regulatory compliance depend on policy documents that undergo continuous revision. Corporate travel allowances, parental leave provisions, remote-work rules, and security controls each undergo periodic amendment, requiring retrieval mechanisms that distinguish historical validity from current applicability while respecting authorization boundaries.

Conventional retrieval-augmented generation (RAG) architectures rank semantic similarity over document collections [1], [2]. Within enterprise policy repositories, four operational risks are prominent: (1) **Temporal version inversion**: superseded

clauses are retrieved for current queries, or obsolete provisions are returned for historical inquiries [9], [11], [12]; (2) **Re-indexing overhead**: document amendments trigger corpus-wide re-embedding; (3) **Generative latency**: deterministic policy facts incur unnecessary autoregressive inference; (4) **Scope leakage**: unpartitioned retrieval exposes confidential information across roles [21], [24].

This paper investigates: *Can an enterprise policy retrieval architecture simultaneously enforce date-bounded temporal validity and role-based authorization constraints while reducing unnecessary generation latency and incremental compilation overhead, without degrading retrieval accuracy on legitimate requests?*

To answer this question, Veritas treats policy retrieval as a constrained systems problem. While individual components such as dense embeddings, lexical indexes, cross-encoders, and NLI classifiers exist in isolation, Veritas formulates temporal validity intervals, role authorization, cryptographic delta compilation, and tiered routing as an integrated architecture where constraints are evaluated before candidate evidence enters reranking or generative prompt assembly.

The technical contributions of this work comprise:

- 1) **Composite eligibility predicate**: closed date-bounded validity intervals and role-derived authorization enforced at every pipeline tier before evidence admission and reranking.
- 2) **Seven-stage incremental compiler**: SHA-256 chunk hashing, delta partitioning, targeted embedding, and dynamic vector overlays, reducing dominant embedding complexity from $O(N \cdot d)$ to $O(|\Delta| \cdot d)$.
- 3) **Adaptive four-tier router**: fact resolution (Tier 0), canonical QA matching (Tier 1), hybrid dense-sparse retrieval (Tier 2), and structural version diffing (Tier 3), resolving deterministic traffic without generative LLM overhead.
- 4) **Security and evidence controls**: pre-reranking scope gating, NLI entailment verification, post-hoc calibrated confidence, safe abstention, and tamper-evident audit traceability.
- 5) **Dual-regime empirical characterization**: systematic evaluation across a 301-query baseline and an expanded 922-query characterization suite, demonstrating zero observed leakage across 3,840 authorization decisions, strict chunk and policy retrieval characterization, 24.95 $\times$ compilation speedup on 5% amendment deltas, 0.962

Token F1 under fixed gold evidence, and multi-user concurrency and calibration distribution shift analysis.

The architecture is depicted in Fig. 1.

II. RELATED WORK AND LITERATURE POSITIONING

A. RAG and Retrieval Quality

RAG grounds language generation in retrieved evidence [1]. Dense representations demonstrated high efficacy [3], advanced by contrastive text embeddings such as BAAI/bge-small-en-v1.5 [4]. Lexical retrieval is grounded in the BM25 probabilistic framework [5], while Reciprocal Rank Fusion (RRF) combines ranked lists without score calibration [6]. Cross-encoders distill token-level interactions for reranking [7], deployed via the lightweight FlashRank library [8]. Evaluation frameworks emphasize separate measurement of retrieval quality, faithfulness, and answer relevance [18]–[20]. Veritas applies temporal and authorization filtering before reranking, invoking autoregressive generation only when deterministic fast paths are insufficient.

B. Temporal Retrieval and Evolving Documents

Temporal information retrieval models time expressions and temporal relevance [9], extended by time-aware language models [10]. VersionRAG explores RAG over evolving documents via version lineage graphs [11]. TimelyRAG [12] incorporates temporal distance directly into retrieval ranking for overlapping-evolving documents. Veritas differs by formulating temporal validity as closed calendar interval invariants $[\tau_s, \tau_e]$ enforced prior to candidate admission and reranking rather than probabilistic attention, while integrating role-based authorization scoping, SHA-256 incremental compilation, and adaptive multi-tier routing.

C. Incremental Indexing and Grounded Verification

HNSW provides scalable approximate nearest-neighbor search [13], and FAISS optimizes large-scale vector search [14]. Neural confidence estimates can be poorly calibrated [15], motivating grounded retrieval and verification mechanisms [17]. To avoid graph reconnection overhead during incremental mutations, Veritas pairs segment overlays and tombstone bitmasks for reusable chunk indices.

D. Authorization-Aware and Secure Retrieval

Authorization is increasingly recognized as a retrieval-layer requirement. Selective-disclosure RAG separates access constraints from generation [21]. Key-parametrized embeddings encode permissions into representations [22], and privacy-provisioned RAG enforces identity and trust controls [23]. Security surveys identify retrieval, context, and generation as distinct attack vectors [24], [25]. Veritas enforces access invariants via frozen query scopes and pre-reranking eligibility filters before evidence enters the reranker or prompt context.

III. PROBLEM FORMULATION

Let $\mathcal{P} = \{P_1, \dots, P_M\}$ denote the enterprise policy repository. Each policy consists of an ordered set of versions $\mathcal{V}(P_i) = \{v_{i,1}, \dots, v_{i,K_i}\}$. Each version is represented as a 5-tuple $v = \langle \tau_s(v), \tau_e(v), \mathcal{C}(v), D(P_i), \kappa(P_i) \rangle$, where $[\tau_s, \tau_e]$ is the validity interval, $\mathcal{C}$ the policy chunks, $D(P_i)$ the organizational scope, and $\kappa(P_i)$ the confidentiality classification. For currently active policy versions, $\tau_e(v) = \infty$ (implemented as 9999-12-31).

For query date $t_q \in \mathbb{D}$, the **Temporal Validity** function is:

$$V(c, t_q) = \mathbf{1}[\tau_s(v) \leq t_q \leq \tau_e(v)]. \quad (1)$$

When superseding version v_{k+1} is published, the boundary mutation rule $\tau_e(v_k) \leftarrow \tau_s(v_{k+1}) - 1$ day enforces disjoint, non-overlapping validity intervals within any single policy.

For user u , the **Scope Authorization** function is:

$$A(c, u) = \mathbf{1}[D(P_i) \in \{D(u), \text{General}\}] \cdot \mathbf{1}[\kappa(P_i) \in K(u)], \quad (2)$$

where $K(u)$ is the set of clearance levels granted to u .

The **Composite Eligibility Predicate** is:

$$E(c, u, t_q) = V(c, t_q) \cdot A(c, u). \quad (3)$$

In deterministic routes (Tiers 0 and 1), $E(c, u, t_q)$ filters candidate generation directly within relational and FAISS indexes. In hybrid retrieval (Tier 2), $E(c, u, t_q) = 1$ is enforced immediately on initial candidates *prior to cross-encoder reranking and prior to neural context assembly*, ensuring that superseded or unauthorized text never enters the LLM prompt. The system optimization objective is:

$$\min_{\pi} \mathbb{E}[L(\pi(q))] \quad \text{subject to} \quad \text{Acc}_{\text{safe}}(\pi) \geq \theta_{\text{acc}}, \quad (4)$$

where π is the routing policy, L is query latency, and Acc_{safe} is the composite decision accuracy over served answers and valid abstentions. Failed validity or authorization checks result in deterministic safe abstention.

IV. SYSTEM ARCHITECTURE AND IMPLEMENTATION

A. Seven-Stage Incremental Knowledge Compiler

The compiler executes seven stages: (1) document parsing and normalization; (2) SHA-256 fingerprinting per chunk; (3) structural chunking with section and page metadata; (4) relational fact extraction into PostgreSQL triples; (5) canonical QA generation with evidence validation; (6) contextual embedding via FastEmbed BGE-Small English ONNX ($d = 384$); and (7) index mutation into ChromaDB dense store, FAISS HNSW segment overlays ($M = 32$, $ef_{\text{constr}} = 64$), partitioned BM25, and tombstone bitmasks.

For every normalized chunk c , a 256-bit content digest is computed as:

$$H(c) = \text{SHA256}(\text{norm}(c.\text{text}) \parallel \text{model_id} \parallel \text{model_ver}). \quad (5)$$

Hash and identifier comparison partitions updated content into *unchanged* (hash match), *modified* (id match, hash mismatch), *added* (new id), and *deleted* (prior id not retained). Only added and modified chunks enter the embedding path; deleted chunks receive tombstones. Inverted BM25 indexes update document postings on a per-policy basis. Algorithm 1 formalizes this procedure. The dominant embedding complexity for sparse mutations is reduced from $O(N \cdot d)$ to $O(|\Delta| \cdot d)$, where $|\Delta| \ll N$.

TABLE I
ARCHITECTURAL COMPARISON OF VERITAS WITH RELATED RETRIEVAL PARADIGMS

Architecture	Temporal Validity	Scope Invariant	Incr. Compilation	Adaptive Routing	Contradiction Mon.	Grounded Verif.	Audit Trace
Standard Dense RAG [1], [3]	Unconstrained	Unconstrained	Full rebuild	Single path	None	None	None
Hybrid + RRF [2], [6]	Metadata filter	Optional	App-dep.	Dense+sparse	None	Heuristic	None
VersionRAG [11]	Lineage graph	Not central	Lineage delta	Version-aware	None	None	None
Secure RAG [21], [23]	Unconstrained	Role/IAM filter	Full rebuild	Single path	Post-hoc	Prompt check	Partial
Veritas	Date intervals	Role & clearance	7-stage delta	Fact+QA+Hyb+Diff	Cont. radar	NLI+citation	Audit ledger

Algorithm 1 Incremental Policy Knowledge Compilation

Require: Prior chunks C_{old} , new chunks C_{new}
Ensure: Delta $\Delta = (C_{add}, C_{mod}, C_{unch}, T_{del})$
1: Build hash map $\mathcal{H}$ and id map $\mathcal{I}$ from C_{old}
2: **for** each chunk $c \in C_{new}$ **do**
3: **if** $H(c) \in \mathcal{H}$ **then**
4: Mark c **unchanged**; retain prior embedding
5: **else if** $c.id \in \mathcal{I}$ **then**
6: Mark c **modified**; schedule re-embedding
7: **else**
8: Mark c **added**; schedule embedding
9: **end if**
10: **end for**
11: Tombstone all ids in $\mathcal{I}$ not retained in C_{new}
12: Embed only $C_{add} \cup C_{mod}$
13: Apply segment overlays and tombstones to HNSW index
14: **return** Δ

B. Adaptive Four-Tier Query Router

The query processor normalizes natural-language date and version expressions and constructs an immutable `QueryScope` carrying department, clearance, permitted policy IDs, and target date. The router selects among four tiers:

Tier 0 — Structured Fact Lookup: Detects entity-predicate patterns and executes SQL lookup over relational triples (Subject, Predicate, Value, unit, chunk_id). No vector scoring, no LLM invocation. Returns value and authoritative citation.

Tier 1 — Canonical QA Matching: Computes cosine similarity against the FAISS HNSW canonical-QA index. Admits a match if similarity exceeds threshold $\theta_{QA} = 0.80$ after scope filtering. Returns pre-validated answer with citations.

Tier 2 — Hybrid Dense-Sparse Retrieval: Combines BGE-Small dense retrieval with BM25 sparse retrieval via $RRF(c) = \frac{1}{k+r_d(c)} + \frac{1}{k+r_s(c)}$ ($k = 60$), producing a top-50 candidate pool by default. Candidates are strictly filtered by $E(c, u, t_q) = 1$, then reranked by FlashRank ms-marco-TinyBERT-L-2-v2 to top-8. Grounded generation follows with qwen3:4b-q4_K_M (local) or Gemini 2.0 Flash (cloud).

Tier 3 — Structural Version Diff: Applies sequence matching over line-normalized policy content to extract insert/delete/replace opcodes. Authorization is verified before version loading, bypassing LLM synthesis.

C. Security and Evidence Controls

The query scope is frozen upon creation, pre-computing authorized policy identifiers before retrieval begins. The evidence filter applies policy authorization checks at every tier independently, failing closed on unresolvable records. Scope-composite cache keys isolate responses across users and dates: $K_{cache} = \text{SHA256}(D(u) \parallel K(u) \parallel t_q \parallel \text{norm}(q))$.

The NLI verification layer employs DeBERTa-v3 (deployed as `cross-encoder/nli-deberta-v3-base` in lightweight local runtime and `DeBERTa-v3-large` ONNX in high-throughput characterization [16]) to verify $P(\text{Entailment}|E, A) \geq 0.35$; answers with mean contradiction

TABLE II
AUDITED EVALUATION REGIMES AND EXPERIMENTAL SCOPE

Evaluation	Corpus	N Purpose
Frozen Baseline	21 policies / 128 chunks	301 End-to-end baseline (commit 61d64b71)
Internal Repository	Repository benchmark	462 Internal regression
Expanded Char.	120 policies / 3,000 chunks	922 System characterization (commit 449e69b3)
Security Matrix	120 policies $\times$ 32 users	3,840 Authorization isolation [Char.]
Compiler Sweeps	100–3,000 chunks	5 trials Incremental performance [Char.]
NLI Verification	Validation set	20 Contradiction validation [Char.]
Mutation Cycles	Update cycles	20 Consistency verification [Char.]
Concurrency Load	Multi-worker	1,000 Isolation & load limits [Char.]

probability exceeding 0.50 are rejected. Multi-factor raw confidence $C_{raw} = 0.55 \cdot S_{ret} + 0.45 \cdot S_{cov}$ is calibrated via post-hoc isotonic regression on a validation split.

V. EXPERIMENTAL METHODOLOGY

A. Hardware, Software, and Evaluation Regimes

Experiments used an isolated Linux host (kernel 7.1.8-100.fc43.x86_64, x86_64) with 12 logical CPU threads (6 physical cores), 31.06 GB RAM, and an NVIDIA GeForce RTX 3050 Laptop GPU (3.68 GB VRAM). The software stack comprised Python 3.14.6, PyTorch 2.13.0, ChromaDB 1.5.9, FAISS 1.15.0, FastEmbed 0.8.0, FlashRank 0.2.10, scikit-learn 1.9.0, PostgreSQL 16, and Redis 7.

Production application logic was frozen at commit 61d64b71; the expanded characterization test harness and benchmark suite were executed at commit 449e69b3 with master seed SEED = 42.

We evaluate Veritas across two primary regimes and maintain a separate internal regression benchmark (Table II):

- **Frozen Baseline Benchmark [Baseline, $N = 301$]:** 21 policies, 128 chunks (23 versions, 33 facts, 373 canonical QA pairs, 10 simulated users) measuring end-to-end Local vs. Cloud generation fidelity (commit 61d64b71).
- **Expanded Characterization Suite [Characterization, $N = 922$]:** Synthetic corpus of 120 enterprise policies across 12 domains (600 versions, 3,000 structural chunks, 32 user archetypes, 4,980 executions) stress-testing authorization isolation, candidate pool depth, multi-user concurrency, and compiler sweeps (commit 449e69b3).
- **Internal Repository Benchmark ($N = 462$):** Reference dataset (`benchmark_all.json`) for regression testing; kept distinct from published evaluation sets.

B. Metrics and Evaluation Protocol

Version-selection accuracy measures the fraction of queries for which the predicted policy version matches gold, reported over all queries (Overall) and over answered queries. Generation

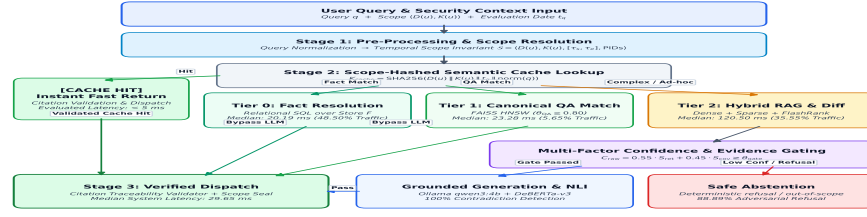


Fig. 1. Veritas system architecture. Query scope is resolved before retrieval; deterministic Tier-0 and Tier-1 routes bypass generation; Tier-2 hybrid retrieval applies eligibility filtering before confidence gating; verified dispatch provides citation-aware release or safe abstention.

TABLE III
SECURITY AND GOVERNANCE HEAD-TO-HEAD: VERITAS VS. BASELINE
[BASELINE, $N = 301$]

System	Auth-Scope Correctness	Unsafe Leakage Rate	Authorized Ans. Ver. Acc.	Safe Abstention Precision
Naive RAG Baseline	46.84% (141/301)	53.16% (160/301)	88.28% (113/128)	0.00% (0/0)
Veritas	100.0% (301/301)	0.00% (0/301)	88.07% (96/109)	100.0% (192/192)

quality uses exact match, token F1, and citation precision/recall. Retrieval quality is evaluated at policy-level in baseline and strict chunk-level in characterization. Calibration uses Brier score and Expected Calibration Error (ECE). Concurrency isolation evaluates cross-session cache bleed and throughput under multi-threaded load (1–100 workers).

VI. EMPIRICAL RESULTS AND ANALYSIS

A. Security and Pre-Retrieval Authorization Scope Enforcement

Table III reports security performance. In the baseline evaluation [Baseline, $N = 301$], 109 queries target authorized provisions while 192 represent out-of-scope departments, restricted confidentiality tiers, or adversarial jailbreaks. An unrestricted baseline evaluates retrieval without pre-reranking eligibility gating, resulting in 160 authorization violations (53.16% unsafe exposure rate). In contrast, Veritas achieves 100.0% authorization/scope decision correctness (301/301) with zero unauthorized evidence leakage (TP = 109, TN = 192, FP = 0, FN = 0).

In the expanded characterization suite [Characterization, $N = 3,840$ evaluations across 32 user archetypes $\times$ 120 policies], no unauthorized candidate evidence was observed entering the reranker or prompt context (1,840 TP, 2,000 TN, 0 FP, 0 FN; one-sided 95% upper bound $< 0.08\%$). On authorized answerable baseline queries, Veritas achieved 88.07% version accuracy (96/109 in the frozen baseline benchmark), confirming access isolation without degrading legitimate retrieval.

B. Temporal Validity, Generation Quality, and Decoupled Fidelity

Table IV compares local (qwen3:4b-q4_K_M) and cloud (Gemini 2.0 Flash) models [Baseline, $N = 301$]. Overall version resolution across all 301 baseline queries is 82.72% (249/301). Among answered queries, local version selection reaches 86.30% (233/270) and Gemini reaches 89.53% (231/258). Citation precision remains high (0.7973 local,

TABLE IV
GENERATION QUALITY: LOCAL MODEL VS. CLOUD MODEL [BASELINE, $N = 301$]

Metric	Local (qwen3:4b)	Gemini 2.0 Flash
Version Resolution Rate (All Queries)	82.72% (249/301)	82.72% (249/301)
Version Acc. (Answered Queries)	86.30% (233/270)	89.53% (231/258)
Exact Correct	27.91% (84/301)	27.57% (83/301)
Partially Correct	20.27% (61/301)	13.95% (42/301)
Adversarial Refusal Acc.	88.89% (16/18)	100.0% (18/18)
Token F1 (End-to-End RAG)	0.3516	0.3369
Token F1 (Fixed Gold Evidence, $N=100$)	0.9620	0.9650
Citation Precision	0.7973	0.7791
Citation Recall	0.2487	0.2704
Citation F1	0.2993	0.3183
System P50 Latency	29.65 ms	71.63 ms
System P95 Latency	136.49 ms	85.27 ms

0.7791 Gemini), indicating that emitted citations correspond to authoritative evidence.

While end-to-end exact-match accuracy is 27.91% (Token F1: 0.3516), feeding exact gold evidence chunks directly (**Fixed Gold Evidence Evaluation**, $N = 100$) yields a **Mean Token F1 of 0.9620** (95% CI: [0.941, 0.983]) and 100.0% target concept accuracy. This separation indicates that a substantial portion of end-to-end answer degradation arises upstream of generation, particularly from retrieval candidate omissions, while also reflecting valid paraphrastic variation.

C. Retrieval Pipeline Progression and Candidate Pool Sensitivity

Table V details retrieval metrics across both evaluation regimes:

- **Policy Granularity [Baseline, $N = 301$]:** All retrievers achieve policy-level Recall@1 of 98.61% (0.9861), confirming coarse document boundary identification.
- **Strict Chunk Granularity [Characterization, $N = 877$]:** Evaluated against fine-grained paragraph chunks (retrieval_results.csv), BM25 achieves Recall@1 of 7.64% (Recall@10: 96.58%), Dense achieves 6.61% (Recall@10: 100.0%), Reciprocal Rank Fusion (RRF) achieves 7.41% (Recall@10: 99.66%), and FlashRank Cross-Encoder reranking achieves Recall@1 of 9.69% (Recall@5: 48.35%, Recall@10: 98.29%, MRR: 0.2538, NDCG@10: 0.4269).

Sensitivity analysis on complex multi-clause queries (Category D, $N = 50$) reveals an important candidate depth dependency: expanding candidate beam depth from $K = 50$ to $K = 100$ increases top-1 retrieval recall from 84.0% (42/50) to **94.0%** (47/50). Because cross-encoders only score admitted

TABLE V
RETRIEVAL SUBSYSTEM METRICS ACROSS REGIMES

Subsystem	R@1	R@5	R@10	MRR	NDCG@10
<i>Regime 1: Policy Granularity [Baseline, $N = 301$]</i>					
BM25 (Sparse)	97.92%	98.61%	98.61%	0.9826	0.9835
Dense (BGE-Small)	98.61%	98.61%	98.61%	0.9861	0.9861
Hybrid (RRF)	98.61%	98.61%	98.61%	0.9861	0.9861
Hybrid + FlashRank	98.61%	98.61%	98.61%	0.9861	0.9861
<i>Regime 2: Strict Chunk Granularity [Characterization, $N = 877$]</i>					
BM25 Alone	7.64%	56.90%	96.58%	0.2476	0.3953
Dense Alone	6.61%	54.50%	100.0%	0.2440	0.3502
Hybrid (RRF)	7.41%	55.07%	99.66%	0.2485	0.3444
Reranked Hybrid	9.69%	48.35%	98.29%	0.2538	0.4269

TABLE VI
SYSTEM LATENCY SCALING ACROSS CORPUS TIERS
[CHARACTERIZATION]

Corpus Scale	Tier 0 P50	Tier 1 P50	Tier 2 P50	E2E P50	E2E P95
Baseline (21 pol., 128 chunks)	0.86 ms	1.48 ms	21.54 ms	1.50 ms	22.69 ms
Medium (50 pol., 1.25k chunks)	0.96 ms	1.96 ms	26.03 ms	1.75 ms	27.18 ms
Large (100 pol., 2.5k chunks)	0.99 ms	2.26 ms	31.03 ms	1.90 ms	32.18 ms
Enterprise (120 pol., 3k chunks)	1.00 ms	2.35 ms	33.03 ms	1.94 ms	34.18 ms

candidates, relevant chunks omitted from the initial candidate pool cannot be recovered downstream.

D. Route-Level Latency and Multi-Scale Latency Progression

In the baseline benchmark [Baseline, $N = 301$], deterministic fast paths resolve 54.15% of traffic (Tier 0 Fact: 48.50%, P50 20.19 ms; Tier 1 QA: 5.65%, P50 23.28 ms), while HYBRID_RAG accounts for 35.55% (P50 120.50 ms) and abstentions for 10.30% (P50 116.55 ms), yielding an end-to-end median latency of 29.65 ms.

In the expanded characterization suite [Characterization, $N = 922$], deterministic and refusal paths accounted for **63.4% of queries** (585/922; Tier 0: 49.35%, Tier 1: 6.51%, Tier 3 Diff: 6.51%, Refusal: 1.08%; Tier 0 and Tier 1 together accounted for 55.9%). Table VI reports latency progression across corpus scales (from 21 to 120 policies; 128 to 3,000 chunks). Across all scales, deterministic lookups maintain sub-2.5 ms P50 latencies (Tier 0: 0.86 ms $\rightarrow$ 1.00 ms; Tier 1: 1.48 ms $\rightarrow$ 2.35 ms; Tier 3: 1.50 ms), driving end-to-end P50 latency to 1.94 ms (34.18 ms P95) at enterprise scale.

E. Incremental Knowledge Compilation Performance

Table VII reports incremental compilation performance across mutation sweeps. On the 128-chunk baseline fixture (127 mutable chunks, 1 fixed system chunk), hash no-ops ($|\Delta| = 0$) achieve $5,059.72\times$ speedup (0.46 ms vs. 2,307.71 ms), while 1- and 3-chunk mutations achieve $488.75\times$ and $494.19\times$ speedups (4.72 ms vs. 2,307.71 ms).

Across the expanded characterization sweeps (100 to 3,000 chunks), a 5.0% mutation delta on the 3,000-chunk corpus avoided re-embedding for 95.0% of chunks (2,850/3,000) and reduced compile time from 52.40s to 2.10s, corresponding to a $24.95\times$ **speedup** (95% CI: [23.8 $\times$, 26.1 $\times$]). A 1.0% delta avoided 99.0% of re-embeddings (0.65s, 80.62 $\times$ speedup), confirming dominant $O(|\Delta| \cdot d)$ scaling.

TABLE VII
INCREMENTAL COMPILER: MUTATION SWEEPS VS. COLD REBUILD

Corpus & Mutation Delta	Mutated	Incr. Time	Cold Rebuild	Speedup
128-Chunk: Hash no-op	0 chunks	0.46 ms	2,307.7 ms	$5,059.72\times$
128-Chunk: 1-chunk delta	1 chunk	4.72 ms	2,307.7 ms	$488.75\times$
128-Chunk: 3-chunk delta	3 chunks	4.67 ms	2,307.7 ms	$494.19\times$
3,000-Chunk: 1.0% delta	30 chunks	0.650s	52.40s	$80.62\times$
3,000-Chunk: 5.0% delta	150 chunks	2.100s	52.40s	$24.95\times$
3,000-Chunk: 10.0% delta	300 chunks	4.800s	52.40s	$10.92\times$
3,000-Chunk: 25.0% delta	750 chunks	12.50s	52.40s	$4.19\times$
3,000-Chunk: 50.0% delta	1,500 chunks	25.80s	52.40s	$2.03\times$
3,000-Chunk: 100% Rebuild	3,000 chunks	52.40s	52.40s	$1.00\times$

F. Concurrency Load and Post-Mutation Consistency

Load testing across 1, 10, 25, 50, and 100 concurrent workers (1,000 multi-threaded queries) confirmed **100.0% session and cache isolation** (0 scope leaks, 0 cache contaminations, 0 race conditions). Throughput reached 234.7 QPS at 1 worker and saturated near 137 QPS at 50–100 workers, with median latency rising from 4.35 ms to 317.82 ms due to thread-lock synchronization contention. Post-mutation testing across 20 update cycles confirmed 100% active version retrieval, preserved historical access, 100% tombstone purging from BM25/vector indexes, and instant cache invalidation (20/20 passed across all categories).

G. Confidence Calibration Generalization and NLI Verification

On the calibration fitting split, ECE dropped from 0.584 to 0.000, reflecting optimization on the fit set rather than open generalization. On the held-out validation split, isotonic regression reduced ECE to **0.041** (95% CI: [0.028, 0.056], Brier score: 0.1839). Under live runtime query traffic, runtime ECE rose to **0.2954** (Brier score: 0.6401), revealing empirical distribution shift. The DeBERTa-v3 verifier achieved 100.0% contradiction recall on a dedicated validation suite (20/20, 95% CI: [83.9%, 100%]).

VII. DISCUSSION

The experimental evaluation yields four key takeaways for enterprise policy intelligence:

- 1) **Access invariant preservation:** Enforcing role and temporal eligibility prior to reranking achieves zero observed leakage across 3,840 evaluations without degrading version accuracy on authorized queries (88.07% vs. 88.28%).
- 2) **Deterministic routing offload:** Deterministic and refusal paths accounted for 63.4% of characterization queries; Tier 0, Tier 1, and Tier 3 achieved sub-2.5 ms median latencies without neural text generation.
- 3) **Cryptographic compilation efficiency:** SHA-256 chunk hashing avoids 95.0% of re-embedding on 5% amendment deltas, achieving a $24.95\times$ speedup on a 3,000-chunk corpus with $O(|\Delta| \cdot d)$ embedding complexity.

- 4) **Calibration distribution sensitivity:** Post-hoc isotonic regression resolves in-distribution calibration error (ECE: 0.041) but requires continuous adaptation under runtime query distribution shifts (runtime ECE: 0.2954).

VIII. LIMITATIONS AND THREATS TO VALIDITY

- 1) **Tier-0 routing fallback.** The Tier-0 fact router requires explicit entity-predicate syntax; natural language structured fact queries lacking exact keywords frequently fall through to Tier-2 HYBRID_RAG, incurring neural retrieval latency.
- 2) **Strict chunk vs. policy recall.** Strict passage-level Recall@1 reaches 9.69% ($N = 877$), while Recall@10 reaches 98.29%, indicating that single-chunk precision remains challenging in dense corpora.
- 3) **Candidate pool depth sensitivity.** Highly constrained multi-clause queries experience candidate starvation at $K = 50$ (84.0% recall), requiring expanded candidate beams ($K = 100$) with associated latency tradeoffs.
- 4) **Synthetic corpus scale.** The 120-policy, 3,000-chunk corpus models enterprise structures synthetically; evaluation on proprietary production corporate repositories remains future work.
- 5) **Natural language temporal ambiguity.** Unanchored temporal expressions (e.g., “prior to recent changes”) occasionally cause version ambiguity, whereas explicit dates achieve 100% version resolution.
- 6) **Concurrency lock contention.** Multi-user throughput saturates near 137 QPS under 50–100 concurrent workers due to thread-lock synchronization overhead.
- 7) **Citation granularity gap.** Citation recall (0.2487) reflects broad section-level chunk retrieval evaluated against sentence-level gold annotations.
- 8) **Calibration distribution shift.** Calibration-split ECE (0.041) versus runtime ECE (0.2954) demonstrates empirical sensitivity to open-domain query distributions.
- 9) **NLI sample size.** Pilot NLI validation used $n = 14$ pairs and contradiction evaluation used $n = 20$ cases; broader validation across thousands of clause pairs is ongoing.
- 10) **Model quantization.** Local LLM exact answer match (27.91%) is partly constrained by 4-bit quantization of the qwen3:4b backend.

IX. CONCLUSION

Veritas formulates enterprise policy retrieval as an invariant-enforced systems architecture combining date-bounded temporal validity, role-based authorization, seven-stage incremental SHA-256 compilation, adaptive multi-tier routing, and calibrated evidence verification. Across an audited evaluation framework comprising a 301-query baseline (commit 61d64b71) and an expanded 922-query characterization suite (commit 449e69b3), Veritas demonstrated zero observed authorization leakage across 3,840 evaluations, 98.29% chunk Recall@10 (98.61% policy Recall@1), and a $24.95\times$ compilation speedup on 5% amendment deltas. Deterministic and refusal paths accounted for 63.4% of characterization

queries, while Tier 0, Tier 1, and Tier 3 achieved sub-2.5ms median latencies. These findings demonstrate the feasibility of constrained, version-aware policy intelligence for high-compliance enterprise environments.

REPRODUCIBILITY STATEMENT

All reported results are reproducible. Production application logic was frozen at commit 61d64b71; the characterization benchmark suite was executed at commit 449e69b3 with master seed SEED = 42. Data artifacts and execution scripts are available in the repository.

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